

Bias/Variance and Ensembling

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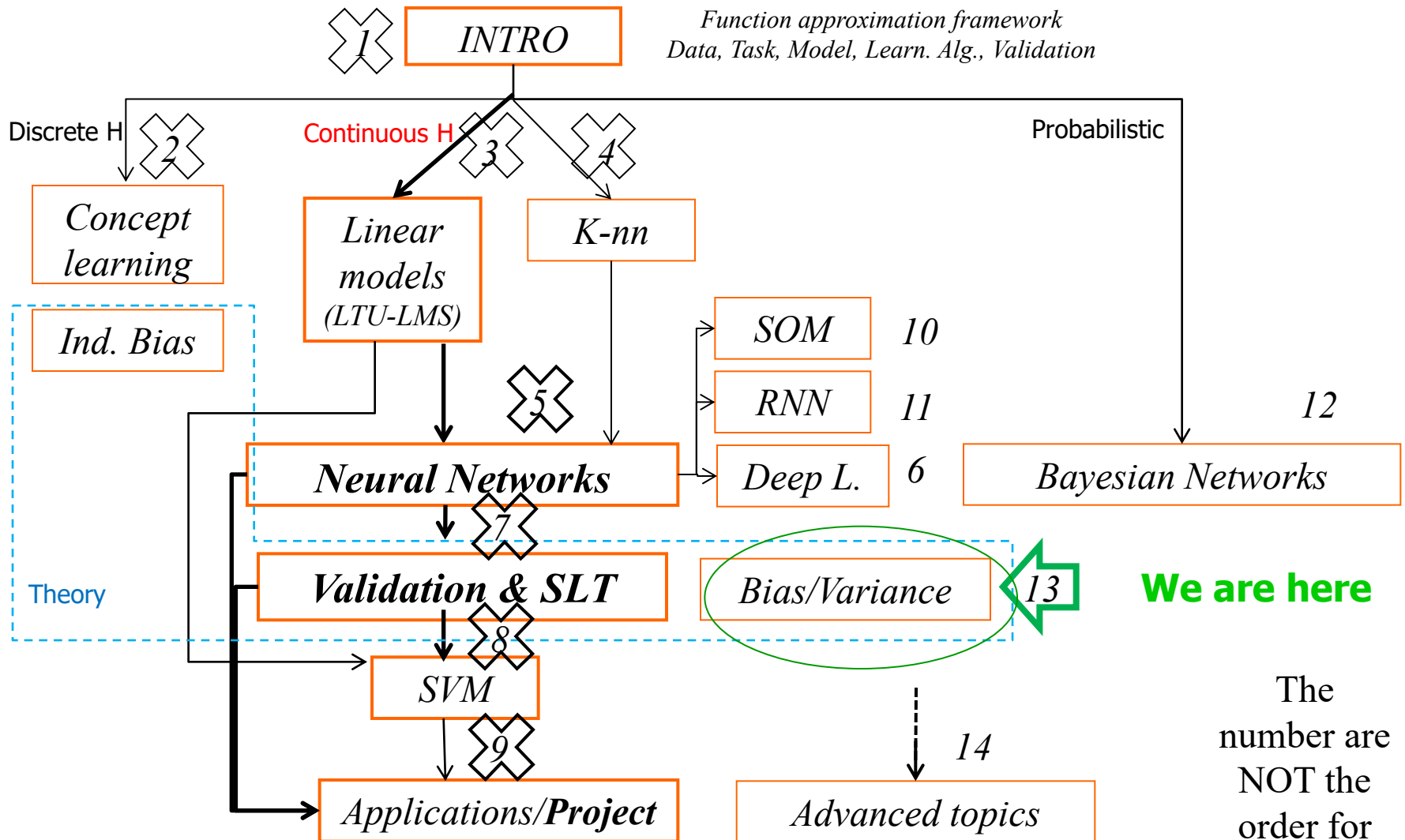
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ML Course structure

Where we are



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Bias-Variance

- A training set is only one possible realization from the universe of data: different training set can provide different estimate
- Decomposition of expected (on various TR) error at a point x :
 - **Bias**: quantifies the discrepancy between true function and $h(x)$ (averaged on data): if H is too small it is high
 - **Variance**: quantifies the variability of the response of model h for different realizations of the training data: due to high flexibility
 - **Noise**: because the label include random error
 - E.g. if for a given x there are more than one possible d
- Assume the **regression** scenario with target y and L_2 (*squared error loss*)

Credits: Pictures from Macskassy, Bishop, Bengio

Bias-Var. Analysis in Regression



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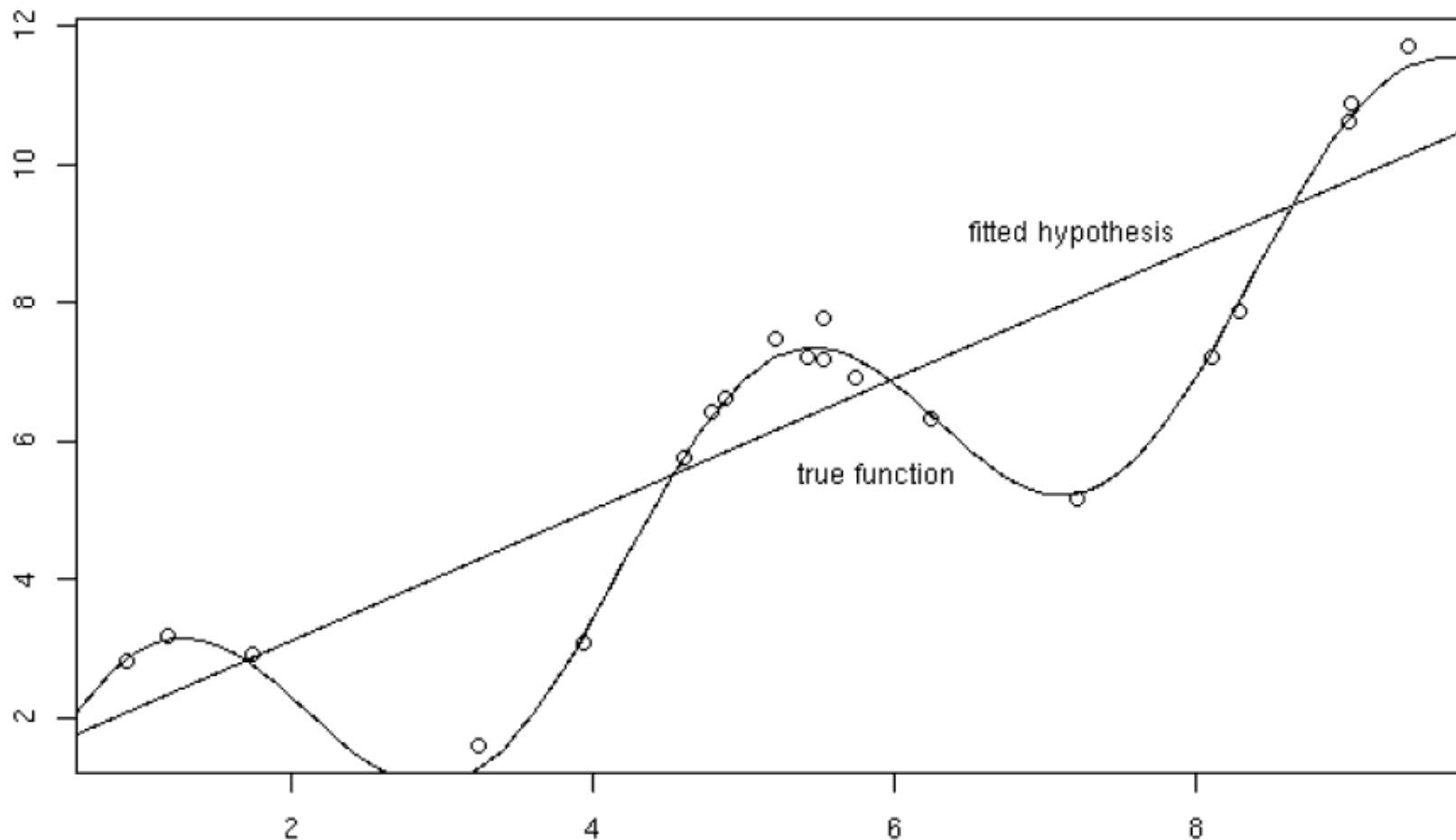
- Suppose we have examples $\langle \mathbf{x}, y \rangle$ where the true function is $y = f(\mathbf{x}) + \varepsilon$ and where ε is Gaussian noise with zero mean and standard deviation σ .
- In linear regression, given a set of examples $\langle \mathbf{x}_i, y_i \rangle \quad i=1..l$ we fit a linear hypothesis $h(\mathbf{x}) = \mathbf{w}\mathbf{x} + w_0$, such as to minimize sum-squared error over the training data:

$$\sum_{i=1}^l (y_i - h(\mathbf{x}_i))^2$$

- 1) Because of the hypothesis class that we chose (linear hypotheses) for some function f , we will have a systematic prediction error
- 2) Depending on the data set we have, the parameters $\mathbf{w}$ that we find will be different.

Example: 20 points

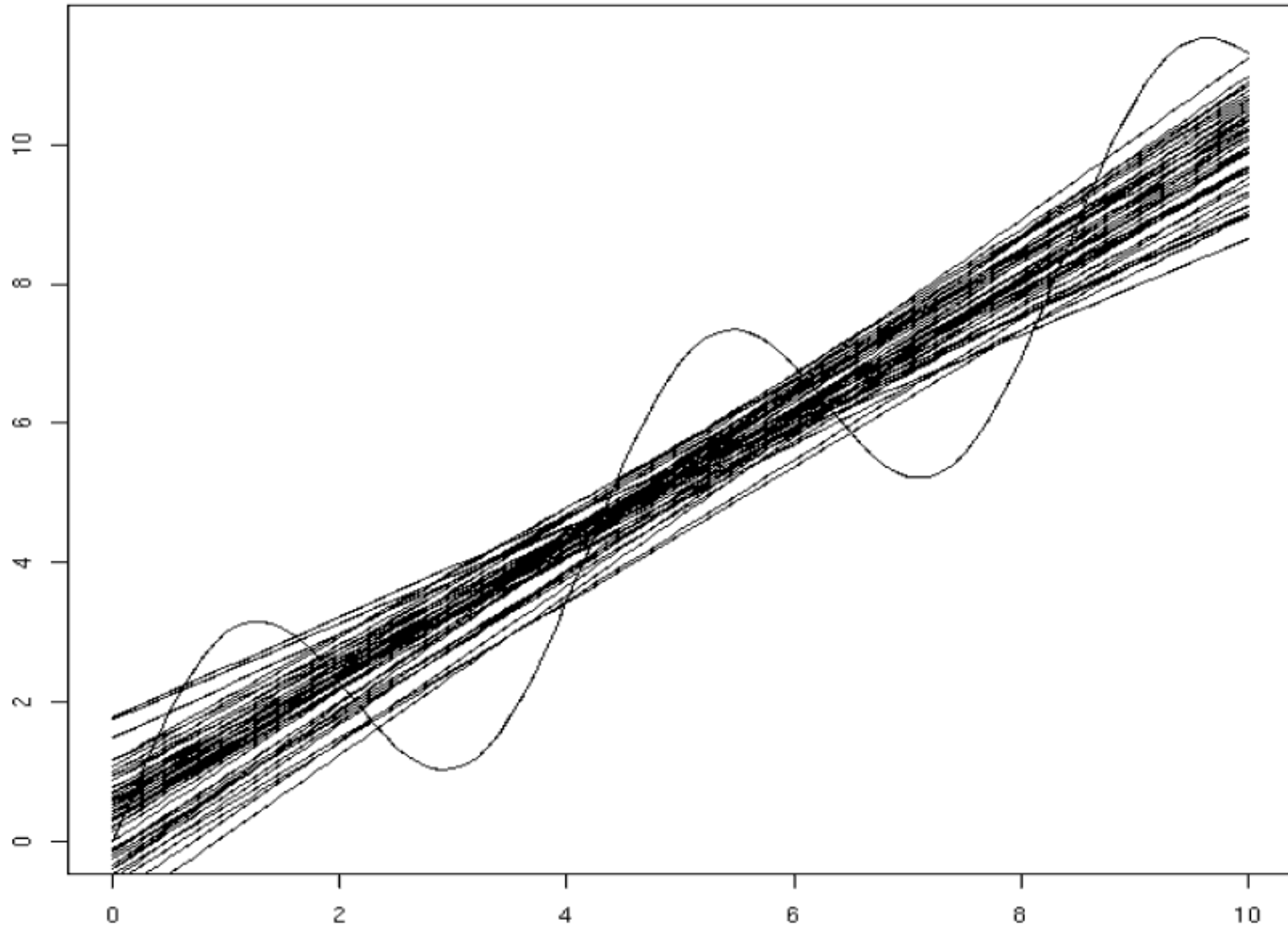
$$y = x + 2 \sin(1.5x) + N(0,0.2)$$



Example: 50 fits (each on 20 points)



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Bias-Var. Analysis

- Given a new data point $\mathbf{x}$, what is the **expected prediction error**?
- Assume that the data points are drawn i.i.d.* from a unique underlying probability distribution P
- The goal of the analysis is to compute, for an arbitrary new point $\mathbf{x}$,

$$E_P \left[(y - h(\mathbf{x}))^2 \right]$$

Note: there is a different h (and y) for each different “extracted” training set :

where y is the value for $\mathbf{x}$ that could be present in a data set, and the expectation is over all training sets drawn according to P .

- We will decompose this expectation into three components:
bias, variance and noise

* i.i.d: independent and identical distributed

Recall of basic statistics

- Let Z be a random variable with possible values z_i , $i=1..l$ and with probability distribution $P(Z)$
- The **expected value** or **mean** of Z is:

$$\bar{Z} = E_P[Z] = \sum_{i=1}^l z_i P(z_i)$$

- If Z is continuous the sum is replaced by an integral, and the distribution by a density function.
- The **variance** of Z is:

$$\begin{aligned} Var[Z] &= E[(Z - \bar{Z})^2] \\ &= E[Z^2] - \bar{Z}^2 \end{aligned}$$

(this last line is proven on the next slide)

Variance Lemma

- $$\begin{aligned} Var[Z] &= E[(Z - \bar{Z})^2] = \sum_{i=1}^l (z_i - \bar{Z})^2 P(z_i) = \\ &= \sum_{i=1}^l (z_i^2 - 2z_i\bar{Z} + \bar{Z}^2) P(z_i) = \\ &= \sum_{i=1}^l z_i^2 P(z_i) - 2\bar{Z} \sum_{i=1}^l z_i P(z_i) + \bar{Z}^2 \sum_{i=1}^l P(z_i) = \\ &= E[Z^2] - 2\bar{Z}\bar{Z} + \bar{Z}^2 \cdot 1 = \\ &= E[Z^2] - \bar{Z}^2 = Var[Z] \end{aligned}$$

- We will use the form

$$E[Z^2] = \bar{Z}^2 + Var[Z]$$

Bias-Var. Decomposition (1)

$$\begin{aligned} E_P \left[(y - h(\mathbf{x}))^2 \right] &= E_P \left[h(\mathbf{x})^2 - 2yh(\mathbf{x}) + y^2 \right] \\ &= E_P \left[h(\mathbf{x})^2 \right] + E_P \left[y^2 \right] - 2E_P[y]E_P[h(\mathbf{x})] \quad * \end{aligned}$$

- Let $\bar{h}(\mathbf{x}) = E_P[h(\mathbf{x})]$ denote the mean prediction of the hypothesis at $\mathbf{x}$, when h is trained with data drawn from P
- For the first term, using the variance lemma*, we have:

$$E_P \left[h(\mathbf{x})^2 \right] = E_P \left[(h(\mathbf{x}) - \bar{h}(\mathbf{x}))^2 \right] + \bar{h}(\mathbf{x})^2$$

- Note that $E_P[y] = E_P[f(\mathbf{x}) + \varepsilon] = f(\mathbf{x})$
- For the second term, using the variance lemma*, we have:

$$E_P \left[y^2 \right] = E_P \left[(y - f(\mathbf{x}))^2 \right] + f(\mathbf{x})^2$$

* Lemma: $E[Z^2] = \text{Var}[Z] + \bar{Z}^2$

Bias-Var. Decomposition (2)

- Putting everything together, we have:

$$\begin{aligned}
 E_P[(y - h(\mathbf{x}))^2] &= E_P[(h(\mathbf{x}) - \bar{h}(\mathbf{x}))^2] + \bar{h}(\mathbf{x})^2 - 2f(\mathbf{x})\bar{h}(\mathbf{x}) + f(\mathbf{x})^2 + E_P[(y - f(\mathbf{x}))^2] \\
 &= E_P[(h(\mathbf{x}) - \bar{h}(\mathbf{x}))^2] + (\bar{h}(\mathbf{x}) - f(\mathbf{x}))^2 + E_P[(y - f(\mathbf{x}))^2] \\
 &= \text{Var}[h(\mathbf{x})] + \text{Bias}[h(\mathbf{x})]^2 + E_P[\varepsilon^2] \\
 &= \text{Var}[h(\mathbf{x})] + \text{Bias}[h(\mathbf{x})]^2 + \sigma^2
 \end{aligned}$$

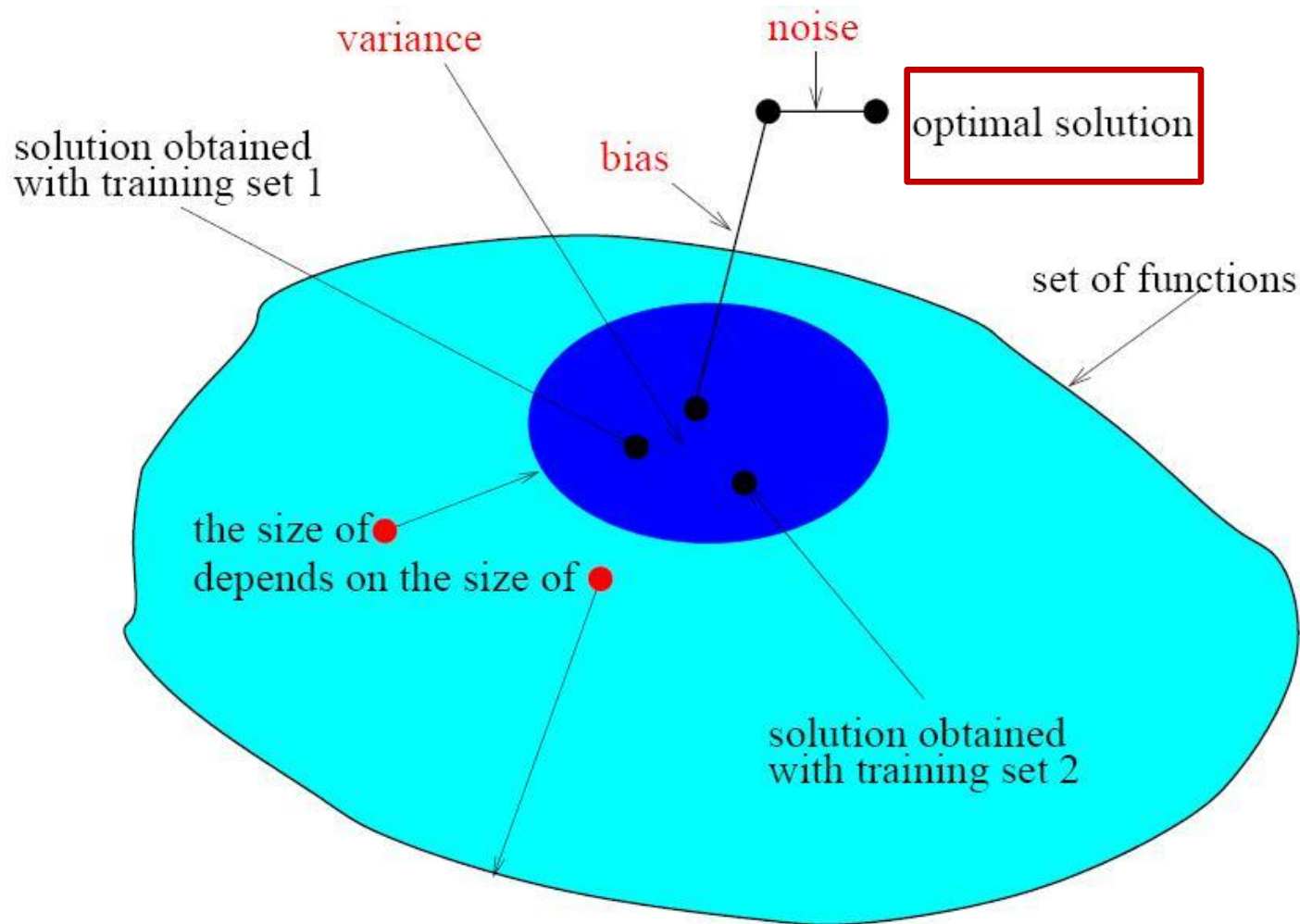
$E_P[h(\mathbf{x})^2] + E_P[y^2] - 2E_P[y]E_P[h(\mathbf{x})]$

- Expected prediction error = Variance + Bias² + Noise²

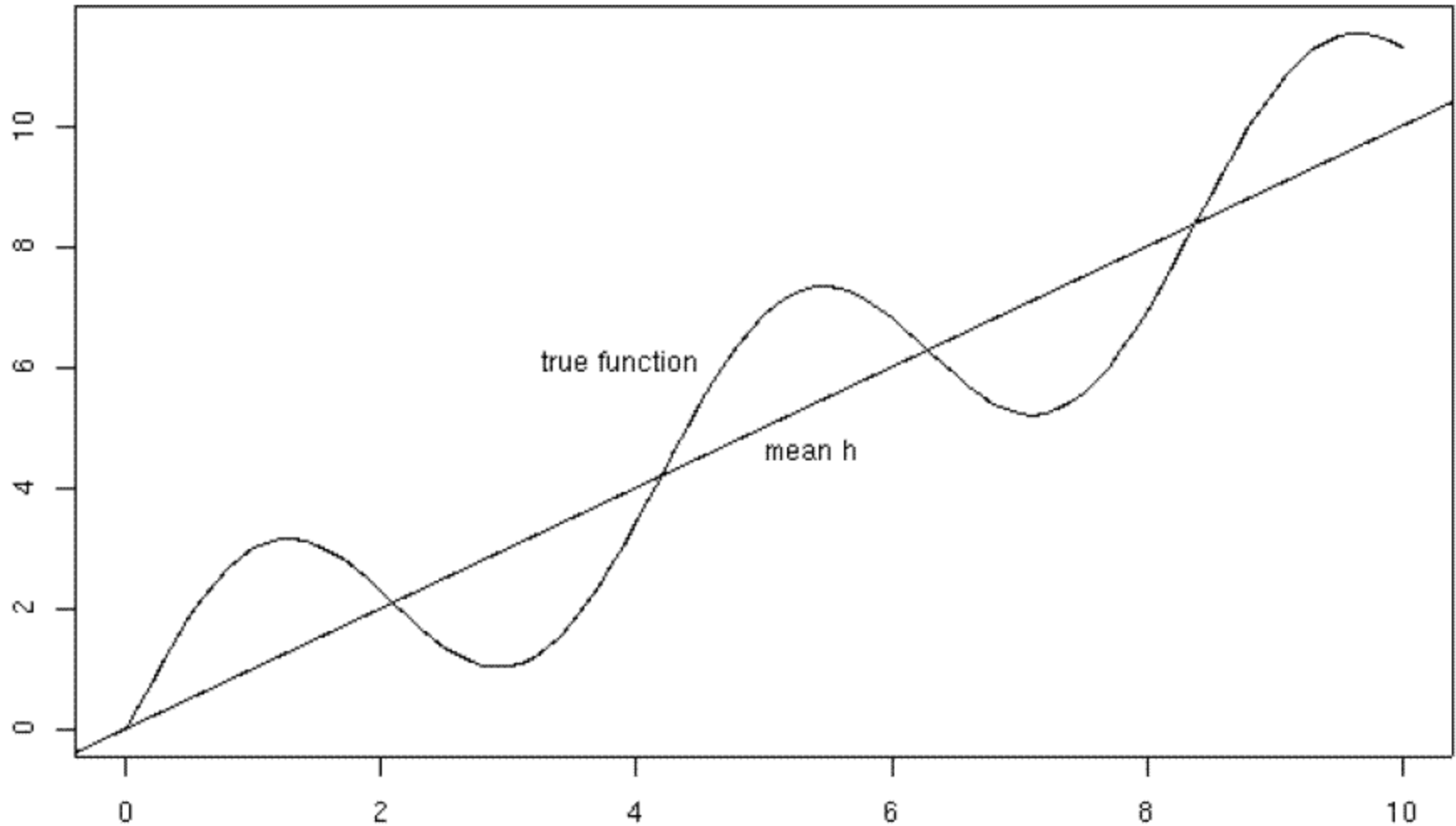
Bias/Variance results

- Decomposition of expected (on various TR) error at a point $\mathbf{x}$:
 - **Bias**: quantifies the discrepancy between true function and $h(\mathbf{x})$ (with $h(\mathbf{x})$ averaged over different TR data) [*systematic error*]
 - E.g. Due to too small H , too rigid model $[\bar{h}(\mathbf{x}) - f(\mathbf{x})]^2$
 - **Variance**: quantifies the variability of the response of model h for different realizations of the training data
 - Due to too high flexibility $E_P [(h(\mathbf{x}) - \bar{h}(\mathbf{x}))^2]$
 - **Noise**: even the optimal solution could be wrong!
 - E.g. if for a given $\mathbf{x}$ there are more than one possible d
[*irreducible*: does not depends on the model,
we have a tolerance on the response (σ , sigma)]

Bias-Variance (graphical view)

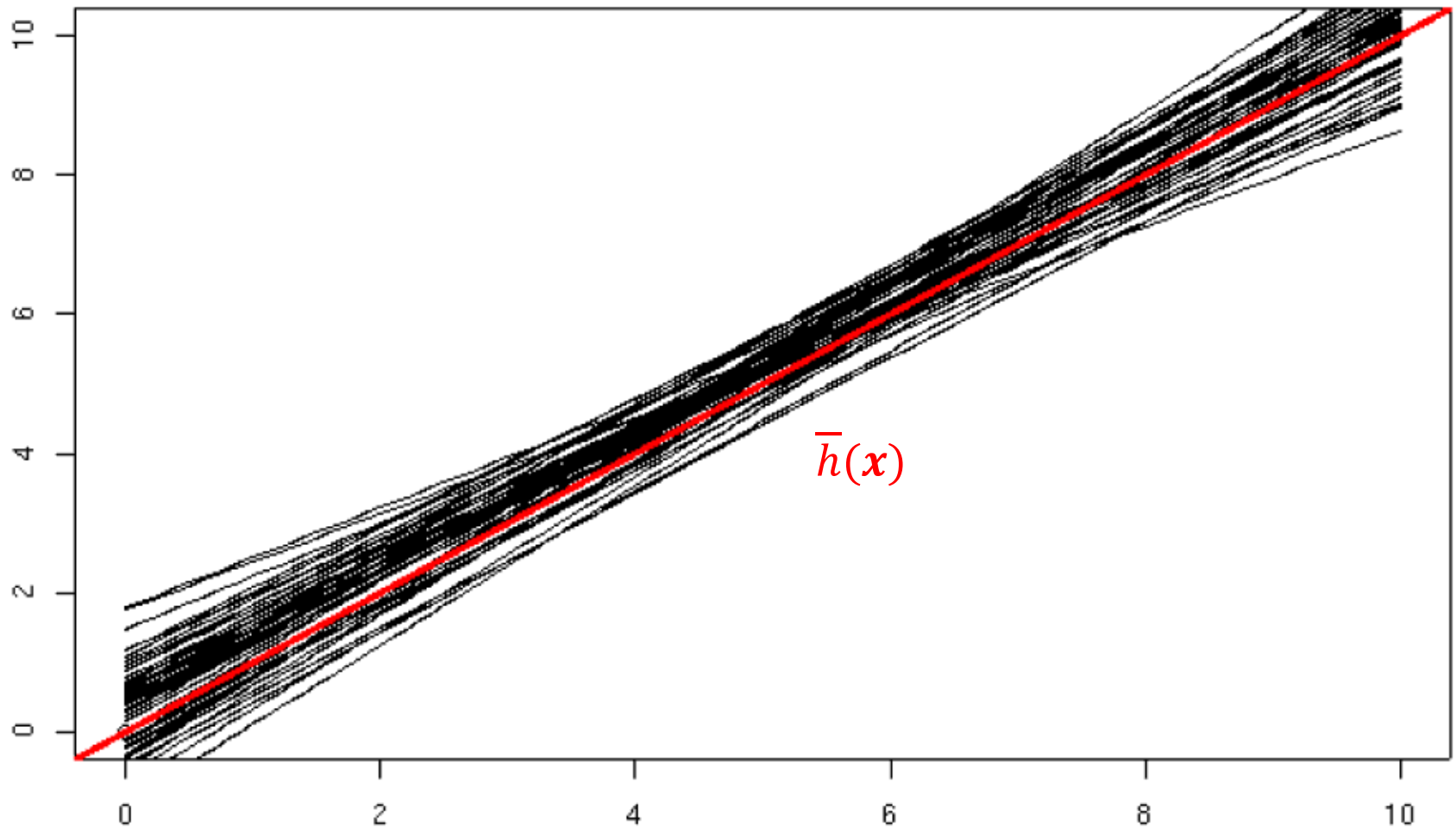


Bias

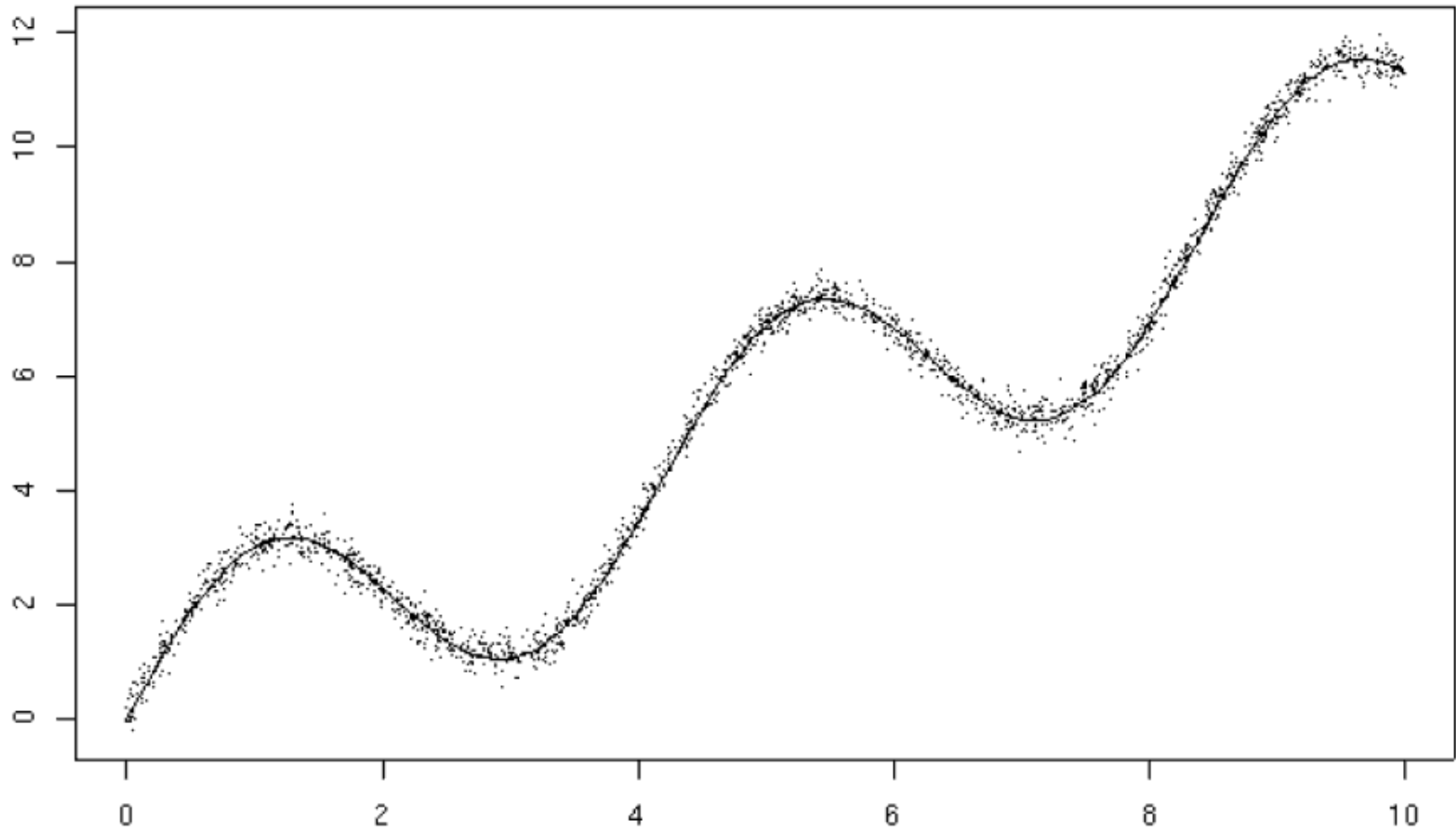


Note the high bias ! (the linear model is too rigid)

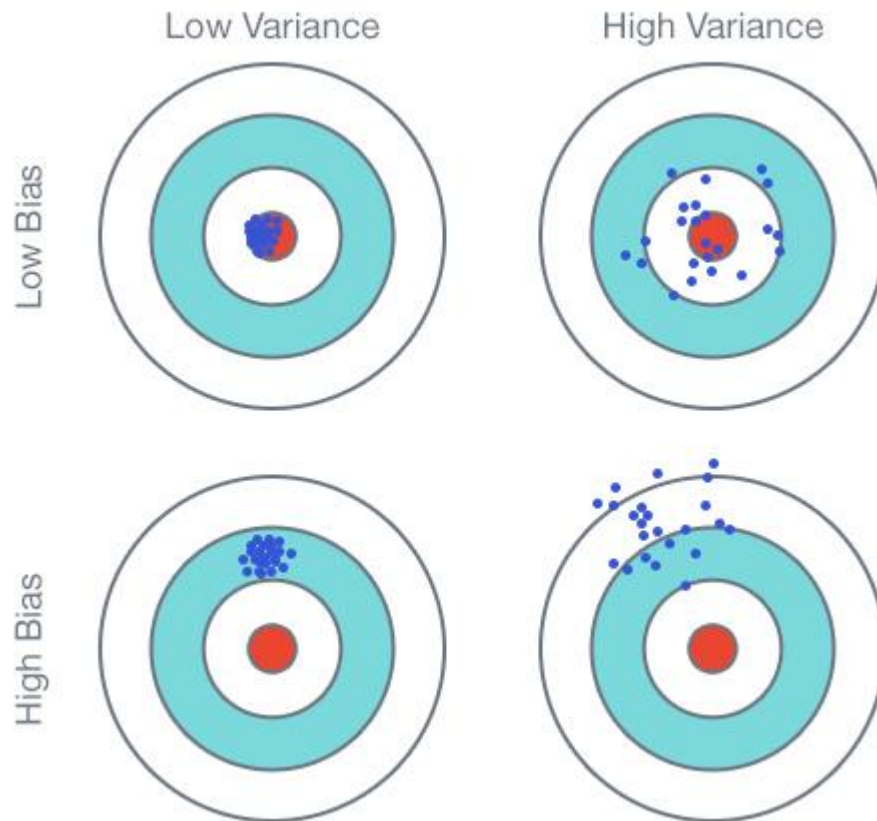
Variance



Noise



A nice graphical analogy for bias and variance



Darts game

Exercise: identify under and overfitting cases...
But the match of the analogy is not perfect for ML...

Bias/Var and Regularization

Recall Regularization:

$$\text{Loss}(\mathbf{w}) = \sum_p (d_p - o(\mathbf{x}_p))^2 + \lambda \|\mathbf{w}\|^2$$

Left (data) term (TR data) Penalty term

Varying λ we can obtain

- less regularized (complex) solutions (low λ)
- more regularized (less complex) solutions (high λ)

- The following example (from Bishop) (should be) using RBF network with LMS

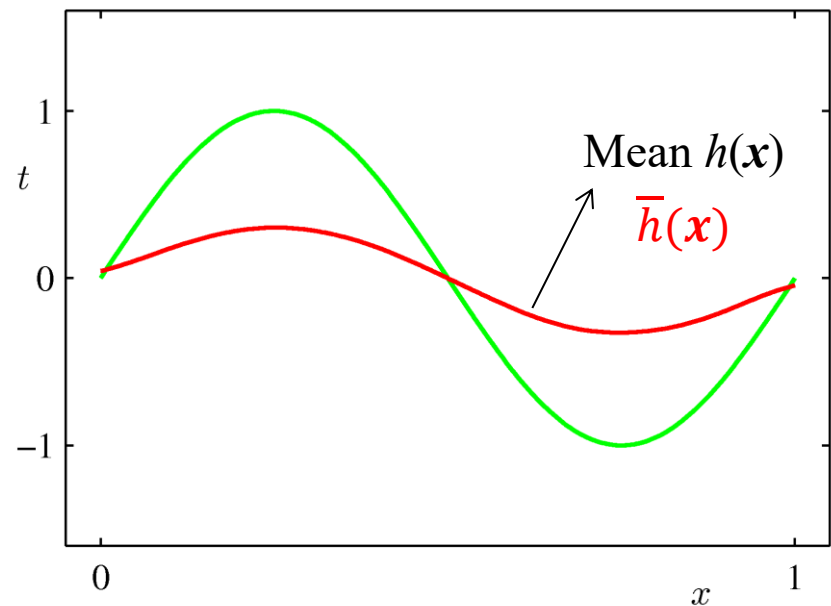
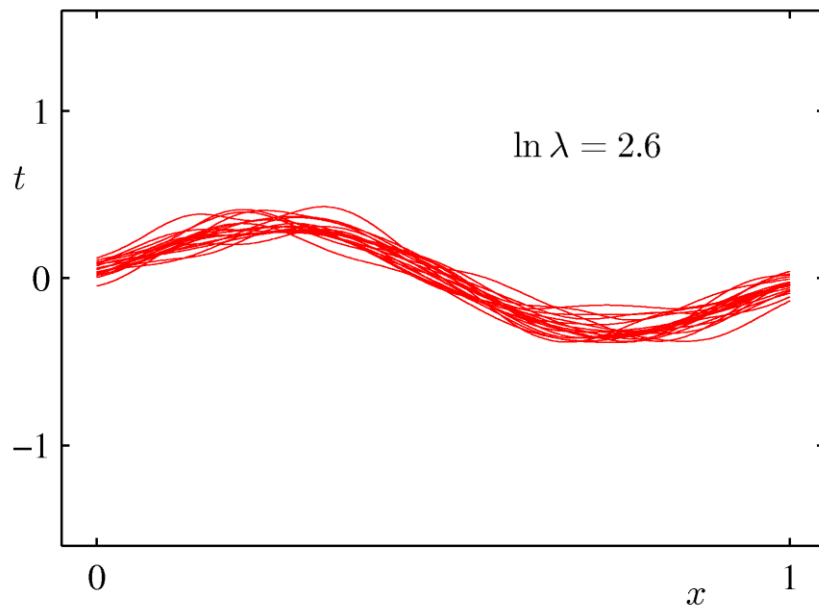
(By the way, note: that the intercept is typically left out of the penalty term)

The Bias-Variance Decomposition: high lambda



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- Example: 25 data sets from the sinusoidal, varying the degree of regularization, lambda (λ)



High bias, low variance

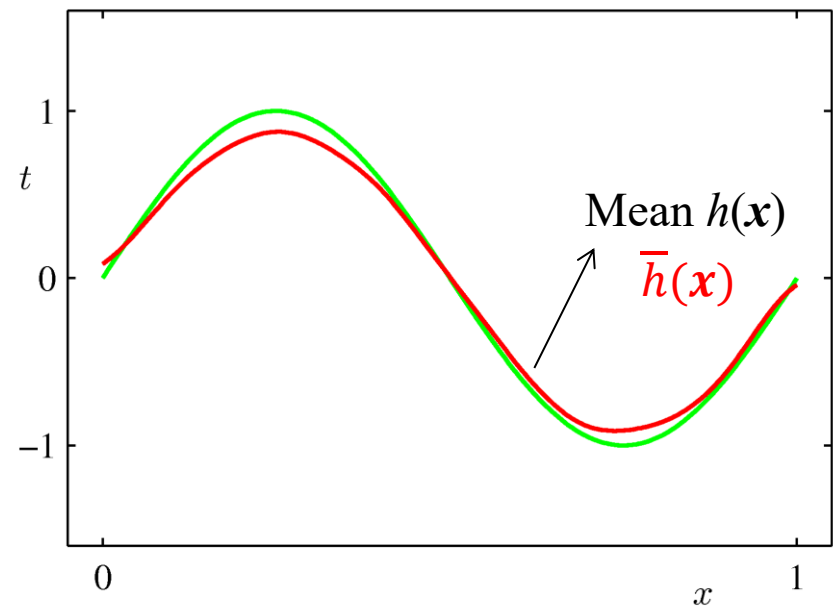
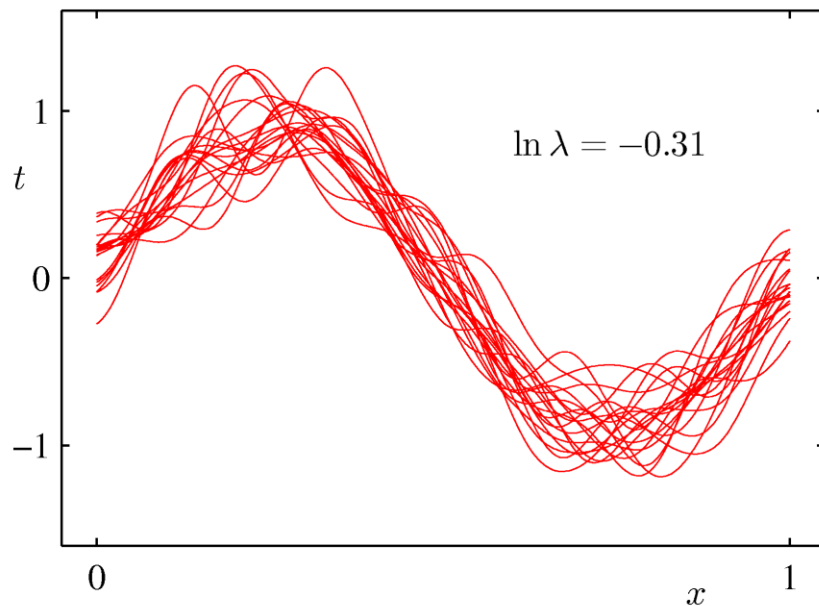
High λ increase the effect of “penalty term”,
less dependent on left (data) term (“too rigid”)

The Bias-Variance Decomposition: medium lambda



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- Example: 25 data sets from the sinusoidal, varying the degree of regularization, lambda (λ).

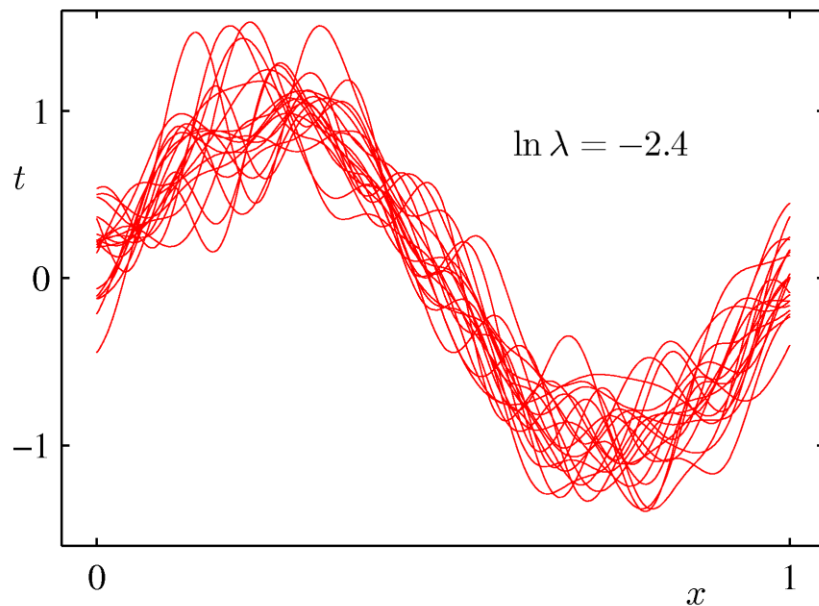


Lower bias, higher variance

Decreasing $\lambda \rightarrow$ we are more dependent
on the specific training data (left –data- term)

The Bias-Variance Decomposition: low lambda

- Example: 25 data sets from the sinusoidal, varying the degree of regularization, lambda (λ).



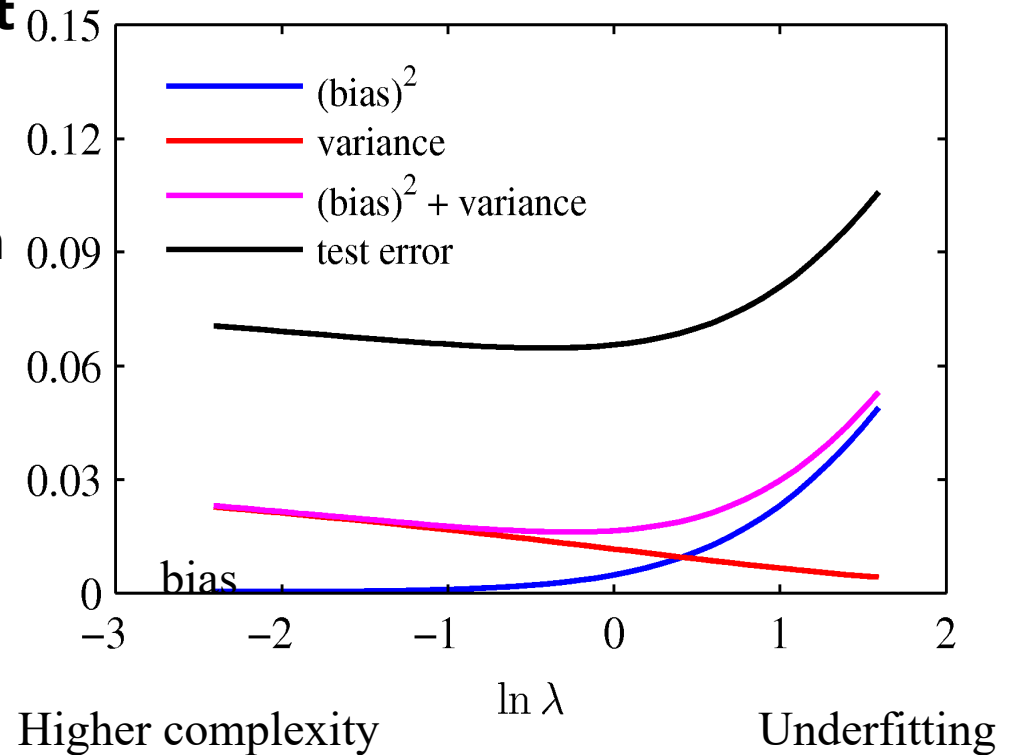
Very low λ : Low bias, high variance:

Test error (for a given TR) can be high (the variance component) but ... even if $h(x)$ are quite variegated, they are correct “on average” (right plot)

The Bias-Variance Trade-off

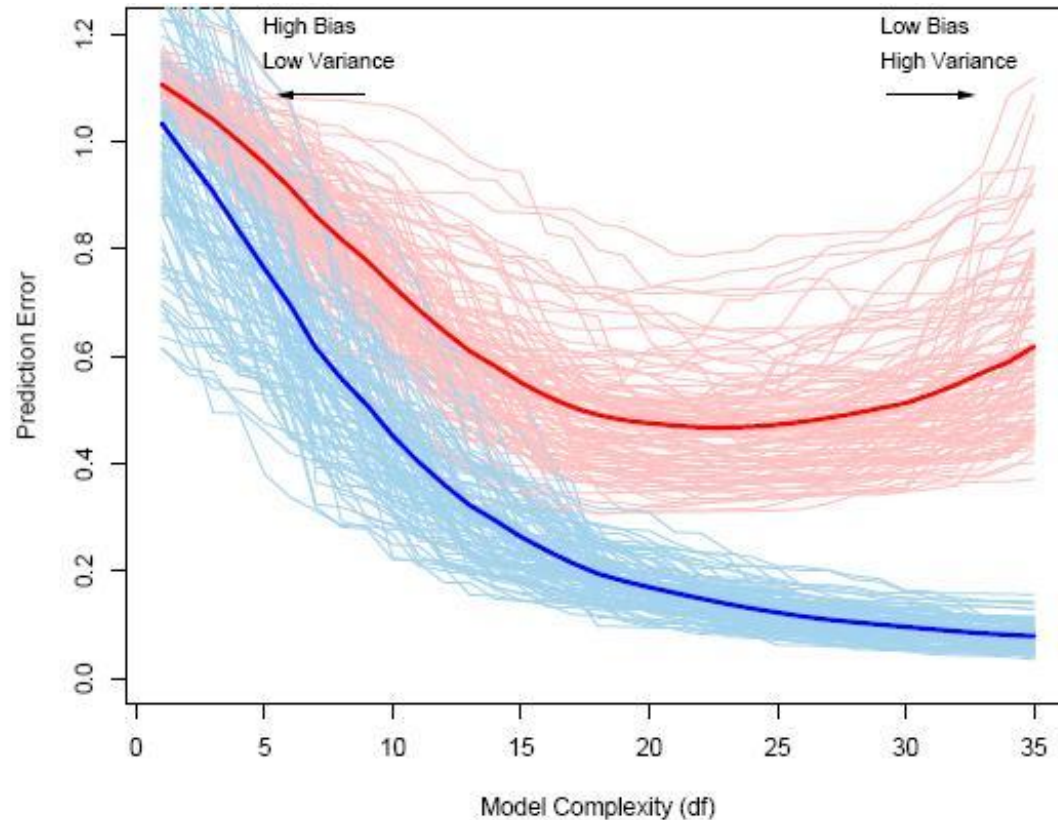
- From these plots, we note that an over-regularized model (large λ) will have a high bias, while an under-regularized model (small λ) will have a high variance.

- Again the trade-off!



Note: complexity is higher on the left
(reverse of our standard plots)

Bias-Variance plot* (with TR and TS)



unlucky

lucky

Complex model : lower
bias, higher variance →
possible overfitting

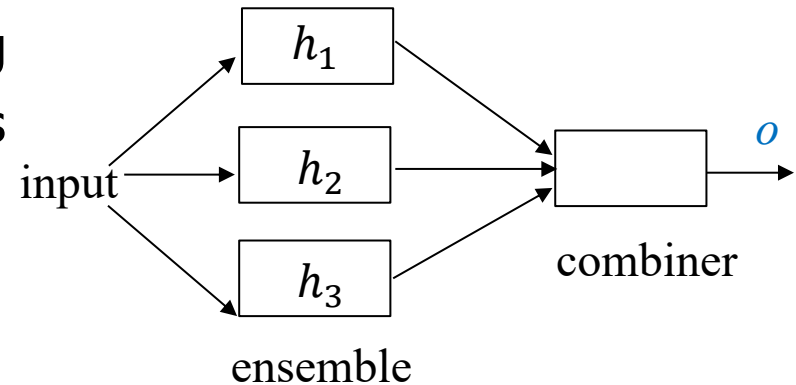
FIGURE 7.1. Behavior of test sample and training sample error as the model complexity is varied. The light blue curves show the training error $\overline{err}$, while the light red curves show the conditional test error Err_T for 100 training sets of size 50 each, as the model complexity is increased. The solid curves show the expected test error Err and the expected training error $E[\overline{err}]$.

Other ML topics

- Ensemble learning: bagging, boosting
- Feature selection
- PAC learning
 - E.g. Mitchell, Haykin
- Comparative analysis of inductive principia (SRM, MDL, Bayesian inference, Regularization) with respect to complexity control
 - Cherkassky, Mulier (chapter 2)

Ensemble Learning

- A first look to **ensemble learning**
- Take advantage of multiple models



Simplest case (“voting schema”):

- Regression
 - simple average committee $o(\mathbf{x}) = 1/K \sum_{i=1}^K h_i(\mathbf{x}) \Rightarrow \bar{h}(\mathbf{x})$
 - $loss(E_i(h_i)) \leq E_i(loss(h_i))$ I.e.
 - (mean_D) square error of a committee $\leq$ mean of the (mean_D) square error
 e.g. $(t-o)^2$, $t=5$, **2 models** $\rightarrow ((5-2)^2 + (5-4)^2)/2 = 5 > (5-3)^2 = 4$
 or even: $((5-6)^2 + (5-4)^2)/2 = 1$ while $(5-5)^2 = 0$
- Classification
 - Take a vote over many classifier (given they provide “diversified” responses): why ? (or classify after the mean of the continuous outputs)

Alternative (“stacking schema”): the combiner is again a ML model

Bagging (*bootstrap aggregating*)

- Combine many classifiers
- Bagging
 - Train K classifiers on different subsets of training set
 - Differentiate each training using bootstrap (resampling with replacement)
- Think to bias-variance (see "*low lambda*" case slide: right figure): $\bar{h}(x)$
- High variance models (with low bias) can perform well *on average*
- Or in other terms: the average of h reduces the variance
 - Regression: use the mean
 - Classification: voting of K classifiers
 - Likely to increase **margin** of a classifier
 - Indeed, average of K separation curves likely to be "in the middle"
"In medio stat virtus"

Boosting (e.g. AdaBoost)

- If models have the same errors we have not advantage on ensembling
- 1. Differentiate each training concentrating on errors (more weight to difficult instances, e.g. more likely to be included in the TR of the next classifier) E.g.:
 - Train 1st classifier
 - Train 2nd classifier, weight more instances misclassified before
 - Etc.
- 2. Combine results by output weights (weighted vote for classifiers, with more weight to low error classifier)
- If not stopped, boosting will learn to classify correctly all instances from training set (using *weak learners* that can merely do better than random guessing, i.e. error less than $\frac{1}{2}$ for binary classification) providing an incremental approach to construct complex models
- Can greatly improve performance of *weak learners*
- Even if bias is reduced the variance does not increase (resist to overfitting)
- As bagging, also likely to produce separation curves *in the middle* (i.e. max margin)
- But can suffer for noisy data (which are weighted more)! (it does not solve the underfitting/overfitting issue in itself)

Feature Selection (few words)

Find a *selection of the features* that are more informative for the problem at hand....

Great benefits for

- Dimensionality reduction (e.g. VC-dim effects for NN, curse of dim, ...), filtering of irrelevant information and noise
(that can lead to great performance improvement)
- Interpretability (e.g. find a set of useful descriptors)
- ❖ It is computationally hard (think how many possible subsets of features can be composed for the learner, and consider the retraining of models)
- ❖ Typically, a heuristic search of the best subset (including or not the learner) by greedy or GA or other optimization techniques.
- ❖ Results are particularly sensible to the validation method: "*feature subset selection bias*", wrong use if selection is made on the entire data set (see slides on CV!)

A survey: I. Guyon, A. Elisseeff. "An Introduction to Variable and Feature Selection". JMLR vol.3:1157-1182, 2003

Bibliography (on bias-variance)



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- Haykin (2nd ed.) sec. 2.13:pp.84-89 : they are not complete !!!
- Haykin (3rd ed.) sec. 2.7:pp.112-116: they are not complete !!!
- Cherkassky, Mulier: 3.4.3
- Hastie, Tibshirani, Friedman : cap 1 e 7

Also ...

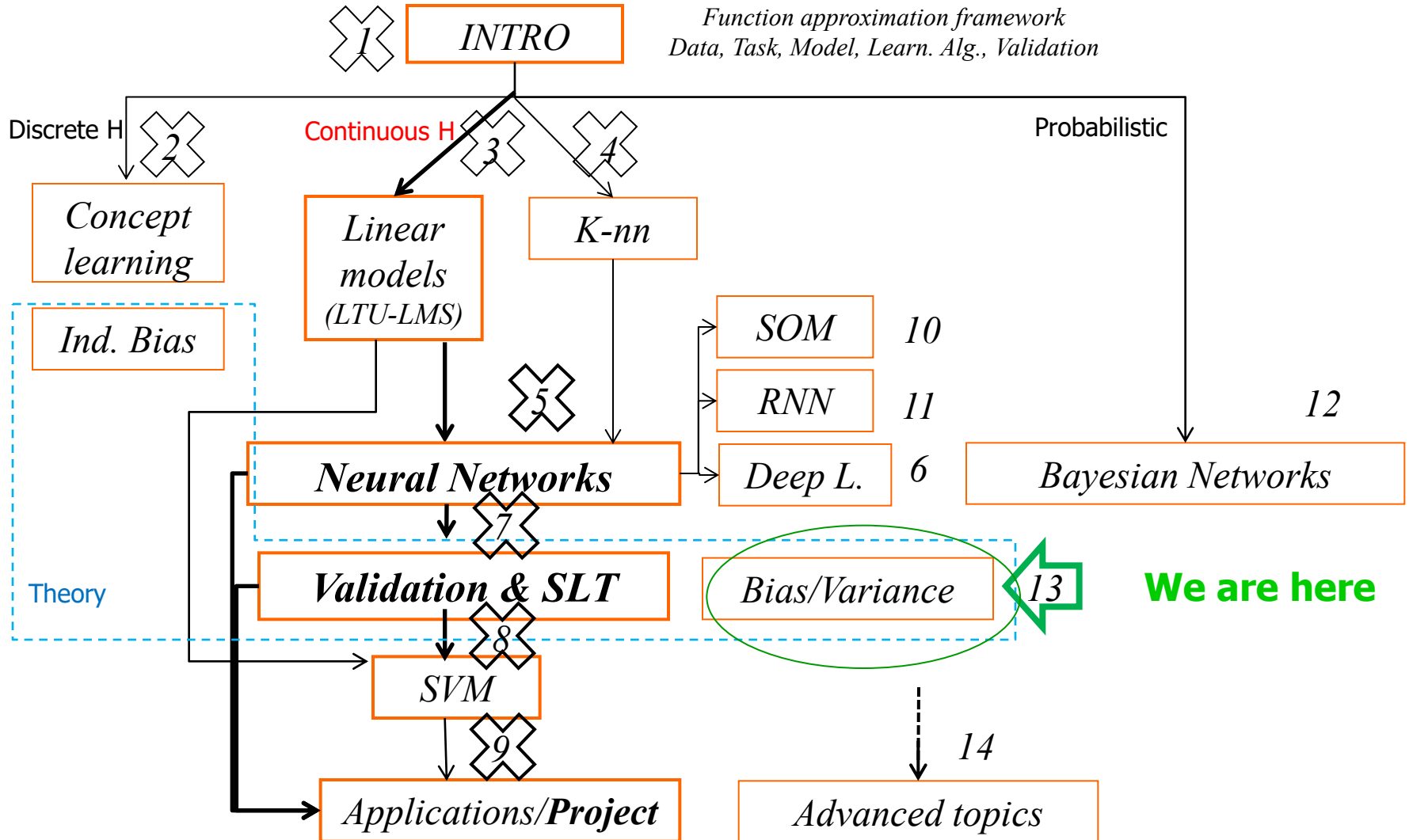
- Bishop,
- Duda, Hart, Stork

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